

Bit Substrings

Time limit: 1.00 s

Memory limit: 512 MB

You are given a bit string of length n . Your task is to calculate for each k between $0 \dots n$ the number of non-empty substrings that contain exactly k ones.

For example, if the string is 101, there are:

- 1 substring that contains 0 ones: 0
- 4 substrings that contain 1 one: 01, 1, 1, 10
- 1 substring that contains 2 ones: 101
- 0 substrings that contain 3 ones

Input

The only input line contains a binary string of length n .

Output

Print $n + 1$ values as specified above.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
101	1 4 1 0